Unit: Unit 4: Absolute Value **Name:** ____________
Topic: Absolute Value Function and its Graph **Date:** ________

Homework (Carolina Reaper)

Q1. The graph of $y = |x - 3|$ has a vertex at
(A) (3, 0)
(B) (0, 3)
(C) (3, 3)
(D) (0, 0)
(E) (5, 0)

Q2. The axis of symmetry for $y = |2x + 1|$ is
(A) $x = -1$
(B) $x = -1/2$
(C) $x = 0$
(D) $x = 1$
(E) $x = 1/2$

Q3. Which of the following is the graph of $y = -|x + 2| + 1$?
(A) A downward-facing V-graph with vertex at (-2, 1)
(B) An upward-facing V-graph with vertex at (-2, -1)
(C) A downward-facing V-graph with vertex at (2, 1)
(D) An upward-facing V-graph with vertex at (2, 1)
(E) A horizontal line at y = 1

Q4. The graph of $y = |x| - 2$ has a minimum value of
(A) -2
(B) -1
(C) 0
(D) 1
(E) 2

Q5. A sound wave has a frequency of $|2x + 1|$ Hz. If $x = -1$, the frequency is
(A) 1 Hz
(B) 2 Hz
(C) 3 Hz
(D) 4 Hz
(E) 5 Hz

Q6. A painter is creating a canvas with a size that is $|x-5|$ feet by $|x+2|$ feet. If $x = 3$, the dimensions are
(A) 2 ft x 5 ft
(B) 3 ft x 4 ft
(C) 4 ft x 3 ft
(D) 5 ft x 2 ft
(E) 6 ft x 1 ft

Q7. A rhythm pattern has a duration of $|3x - 2|$ seconds. If $x = 1$, the duration is
(A) 1 s
(B) 2 s
(C) 3 s
(D) 4 s
(E) 5 s

Q8. The graph of $y = |x + 1| - 3$ has a maximum value of
(A) -3
(B) -2
(C) -1
(D) 0
(E) 1

Open Ended Questions (FRQ)

Q9. Graph the function $y = |2x - 1| + 2$ and identify its vertex, axis of symmetry, and direction of opening. Required space: 5 lines

Q10. A music composer is creating a song with a frequency that follows the function $y = |x + 2| - 1$. If the frequency at $x = 0$ is 1 Hz, find the value of x when the frequency is 3 Hz. Required space: 5 lines

Chef's Tips

- Tip 1: Emphasize the importance of identifying the vertex, axis of symmetry, and direction of opening for absolute value graphs
- Tip 2: Encourage students to analyze the given functions and identify any transformations
- Tip 3: Remind students to use the properties of absolute value functions to solve problems